Reg. No. | | | | | | | | | | |

[13EE7609/12EE7609]

SEVENTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, DECEMBER – 2016

OPERATIONS RESEARCH

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. Define Operations Research. List and explain the various phases of an operations research study. **07**

b. What are the different characteristics of OR? Explain each one in detail. **07**

2. a. A manufacturer produces two types of models A and B. Each A model requires 4 hours of grinding and 2 hours of polishing, whereas each B model requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and each polisher works for 60 hours a week. Profit on an A model is Rs 3 and on B model is Rs 4. Whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week? Formulate this as an LPP and solve graphically. **08**

b. Solve the following LPP by simplex method **06**

Maximize $Z = 4x_1 + 10x_2$

Subject to constraints

$2x_1 + x_2 \leq 50$

$2x_1 + 5x_2 \leq 100$

$2x_1 + 3x_2 \leq 90$

$x_1, x_2 \geq 0$

3. a. Explain the concept of duality in linear programming. Also explain the rules for converting primal problem into its dual. **06**

b. Write the dual for the following primal **08**

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to constraints

$x_1 + 2x_2 + 3x_3 = 15$

$2x_1 + x_2 + 5x_3 - 3x_4 \geq 20$

$x_1 + 2x_2 + x_3 + x_4 \leq 10$

$x_1, x_4 \geq 0$; x_2, x_3 are unrestricted.

4. a. A product is produced by four factories A, B, C and D. the unit production costs in them are Rs 2, Rs 3, Re 1 and Rs 5 respectively. Their production capacities are: factory A – 50 units, B – 70 units, C – 30 units and D – 50 units. These factories supply the product to four stores, demands of which are 25, 35, 105 and 20 units respectively. Unit transportation cost in Rs from each factory to each store is given in the table below. Determine the extent of deliveries from each of the factories to each of the stores so that the total production and transportation cost is minimum. **08**

		Stores			
		1	2	3	4
Factories	A	2	4	6	11
	B	10	8	7	5
	C	13	3	9	12
	D	4	6	8	3

b. Solve the following transportation problem; the matrix below represents the time. Obtain the minimum transportation time. **06**

		To				Available
		D1	D2	D3	D4	
From	O1	10	0	20	11	15
	O2	1	7	9	20	25
	O3	12	14	16	18	5
	Required	12	8	15	10	

PART - B

5. a. A travelling salesman has to visit 5 cities. He wishes to start from a particular city, visit each city once and then return to his starting point. Cost of going from one city to another is shown below. Find the least cost route. **08**

		To city				
		1	2	3	4	5
From city	1	∞	10	25	25	10
	2	1	∞	10	15	2
	3	8	9	∞	20	10
	4	14	10	24	∞	15
	5	10	8	25	27	∞

b. A small garment making unit has five tailors stitching five different types of garments. All the five tailors are capable of stitching all the five types of garments. The output per day per tailor and the profit (Rs) for each type of garment are given below. Which type of garment should be assigned to which tailor in order to maximize profit, assuming that there are no constraints? **06**

		Garments				
		1	2	3	4	5
Tailors	A	7	9	4	8	6
	B	4	9	5	7	8
	C	8	5	2	9	8
	D	6	5	8	10	10
	E	7	8	10	9	9
Profit (Rs) per garment		2	3	2	3	4

6. a. Define Game Theory. Explain in detail the characteristics of game theory. **07**

b. Use dominance method to reduce the following game to either (2 X n) or (m X 2) game and then solve it graphically **07**

		Player B			
		I	II	III	IV
Player A	I	19	6	7	5
	II	7	3	14	6
	III	12	8	18	4
	IV	8	7	13	-1

7. a. The following table gives data on normal time and cost and crash time and cost for a project. **08**

Activity	Normal		Crash	
	Time (days)	Cost (Rs)	Time (days)	Cost (Rs)
1 – 2	8	100	6	200
1 – 3	4	150	2	350
2 – 4	2	50	1	90
2 – 5	10	100	5	400
3 – 4	5	100	1	200
4 – 5	3	80	1	100

If overhead cost is Rs 80/ day

i. What is the normal duration and its cost?

ii. What is the lowest cost schedule and its cost?

iii. What is the minimum length and its cost?

b. The following table shows the jobs of a PERT network with their time estimate in days. **06**

Job	Duration (days)		
	Optimistic	Most likely	Pessimistic
1 – 2	3	6	15
1 – 6	2	5	14
2 – 3	6	12	30
2 – 4	2	5	8
3 – 5	5	11	17
4 – 5	3	6	15
6 – 7	3	9	27
5 – 8	1	4	7
7 - 8	4	10	28

i) Draw the network diagram
ii) find the project length and variance of the project
iii) If the due date is 43 days, what is the probability of project completion.

8. a. What do you mean by Replacement analysis? What are the causes of replacement? Also explain the need for replacement. **06**

b. A truck owner from his past experience estimated that the maintenance cost per year of a truck whose purchase price is Rs 1, 50, 000 & the resale value of the truck will be as follows: **08**

Year	1	2	3	4	5	6	7	8
Maintenance cost (Rs in 1000)	10	50	20	25	30	40	45	50
Resale value (Rs in 1000)	130	120	115	105	90	75	60	50

Determine at which time it is profitable to replace the truck.